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Link GitHub : https://github.com/salmn-tkyk/JurnalModul8_KPL.git

Laporan Jurnal Modul 8

Output:

```
Microsoft Visual Studio Debu

Please insert the amount of money to transfer: 50000000
Transfer fee = 15000 Total amount = 50015000
Select transfer method: 1. RTO(real - time)
2. SKN
3. RTGS
4. BI FAST
4
Please type yes to confirm the transaction:
yes
The transfer is completed
```

```
4 references
public class BanktransferConfig
{
    8 references
    public string lang { get; set; }
    4 references
    public TransferConfig transfer { get; set; }
    3 references
    public List<string> methods { get; set; }
    3 references
    public ConfirmationConfig confirmation { get; set; }

    1 reference
    public BanktransferConfig()
    {
        lang = "en";
        transfer = new TransferConfig(25000000, 6500, 15000);
        confirmation = new ConfirmationConfig("yes", "ya");
        methods = new List<string> { "RTO(real - time)", "SKN", "RTGS", "BI FAST" };
    }
}

4 references
public class TransferConfig
{
    2 references
    public string threshold { get; set; }
    2 references
    public string low_fee { get; set; }
    2 references
    public string high_fee { get; set; }

    [JsonConstructor]
    0 references
    public TransferConfig() { }
    1 reference
    public TransferConfig(int threshold, int low_fee, int high_fee)
    {
        this.threshold = threshold.ToString();
        this.low_fee = low_fee.ToString();
        this.high_fee = high_fee.ToString();
    }
}
```

```
public class ConfirmationConfig
{
    2 references
    public string en { get; set; }
    2 references
    public string id { get; set; }

    [JsonConstructor]
    0 references
    public ConfirmationConfig() { }
    1 reference
    public ConfirmationConfig(string en, string id)
    {
        this.en = en;
        this.id = id;
    }
}
```

BankTransferConfig berfungsi sebagai kontainer utama yang menyimpan preferensi bahasa (lang), detail biaya transfer (transfer), daftar metode transaksi (methods), serta pesan konfirmasi dalam berbagai bahasa (confirmation). Di dalam konstruktornya, aplikasi secara otomatis menetapkan nilai default, seperti bahasa Inggris ("en"), ambang batas transfer Rp25.000.000 dengan biaya rendah Rp6.500 dan biaya tinggi Rp15.000, dan menyediakan pilihan metode transfer seperti RTO, SKN, RTGS, dan BI FAST.

```
class Program
{
    0 references
    static void Main(string[] args)
    {
        string filePath = "bank_transfer_config.json";
        BanktransferConfig config;

        if (File.Exists(filePath))
        {
            string jsonString = File.ReadAllText(filePath);
            config = JsonSerializer.Deserialize<BanktransferConfig>(jsonString);
        }
        else
        {
            config = new BanktransferConfig();
            string jsonString = JsonSerializer.Serialize(config, new JsonSerializerOptions { WriteIndented = true });
            File.WriteAllText(filePath, jsonString);
        }

        if (config.lang == "en") Console.Write("Please insert the amount of money to transfer: ");
        else Console.Write("Masukkan jumlah uang yang akan di-transfer: ");

        int inputTf = int.Parse(Console.ReadLine());

        int biayaTf = inputTf <= int.Parse(config.transfer.threshold) ? int.Parse(config.transfer.low_fee) : int.Parse(config.transfer.high_fee);
        int total = inputTf + biayaTf;

        if (config.lang == "en") Console.WriteLine($"Transfer fee = {biayaTf} Total amount = {total}");
        else Console.WriteLine($"Biaya transfer = {biayaTf} Total biaya = {total}");

        if (config.lang == "en") Console.Write("Select transfer method: ");
        else Console.Write("Pilih metode transfer: ");
```

```
for (int i = 0; i < config.methods.Count; i++)
{
    Console.WriteLine($"{i + 1}. {config.methods[i]}");
}
string method = Console.ReadLine();

string confirmToken = config.lang == "en" ? config.confirmation.en : config.confirmation.id;
if (config.lang == "en") Console.WriteLine($"Please type {confirmToken} to confirm the transaction:");
else Console.WriteLine($"Ketik {confirmToken} untuk mengkonfirmasi transaksi:");

string inputConfirm = Console.ReadLine();

if (inputConfirm == confirmToken)
{
    Console.WriteLine(config.lang == "en" ? "The transfer is completed" : "Proses transfer berhasil");
}
else
{
    Console.WriteLine(config.lang == "en" ? "Transfer is cancelled" : "Transfer dibatalkan");
}
```

Kode ini akan membaca file bank_transfer.config.json, jika file ditemukan, program memuat pengaturannya, jika tidak, program akan membuat file konfigurasi baru dengan nilai default. Setelah itu, program meminta input jumlah uang dari pengguna dan menghitung biaya transfer berdasarkan batas yang sudah ditentukan dalam konfigurasi

```
<No Schema Selected>
{
  "lang": "CONFIG1",
  "transfer": {
    "threshold": "CONFIG2",
    "low_fee": "CONFIG3",
    "high_fee": "CONFIG4"
  },
  "methods": "CONFIG5",
  "confirmation": {
    "en": "CONFIG6",
    "id": "CONFIG7"

}
```

Format JSON yang digunakan untuk menyimpan pengaturan aplikasi.